

Introduction to Molecular Biology

1. Use the Codon Table to find the mRNA and gene that might translate to the following protein: VKLFPWFNQY
 - a. How many possibilities are there for this gene/mRNA?
2. What is the reverse complement of the following dna sequence: CTGGTCAGGTCA? It is a standard to write the DNA sequence from 5' → 3'. Write your reverse complement following that.
3. In general, the genes or coding regions of the DNA to be transcribed to mRNA and then translated into proteins. The number of genes in an organism is small compared to the number of proteins. How are these large numbers of proteins then translated?
4. What are the fundamental tasks of the following entities?
 - a. DNA Polymerase
 - b. RNA polymerase
 - c. mRNA
 - d. tRNA
 - e. sRNA
 - f. Ribosome
 - g. Proteins
 - h. Transcription Factors
5. How many k-mers are possible for k=4 for DNA, RNA and Protein sequences?

Finding Origin of Replication

1. Draw the skew diagram for the following genome:
CGGCAGCGTCGCCACACGATCGATCGCGTCGTCGATGCATGCTA
2. What is the runtime of the these methods shown in class:
 - a. PatternCount(Text, Pattern)
 - b. FrequentWords(Text, k)
3. Find out all k-mers with 2 or less hamming distance from the following k-mer: ACGT
4. What role does the directionality of DNA play in finding origin or replication?

Finding Regulatory Motifs

1. Consider the following set of motifs: ACGT, ACCT,AGGT, ATAC, ACGG. Now answer the following:

ACGT
ACCT
AGGT
ATAC
ACGG

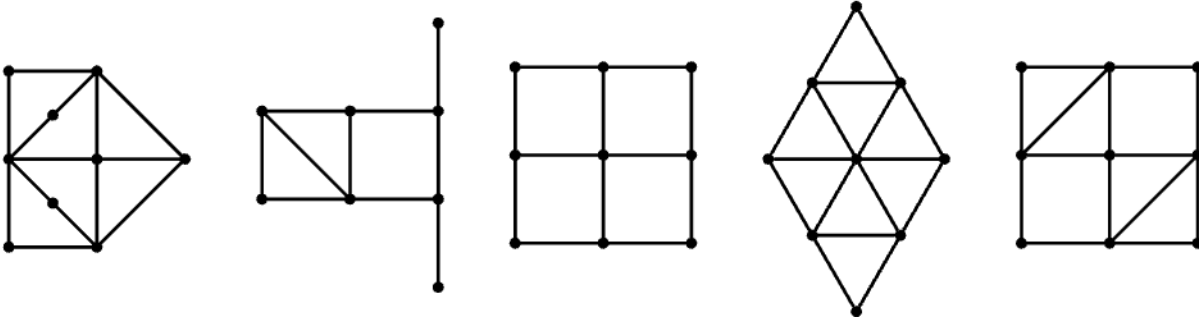
- a. What is the score of this motif set if we use the hamming distance version?
- b. Calculate the profile matrix for this set.

- c. What is the score of this motif set if we use the entropy version? (use laplacian correction, constant = 1)
 - d. If we have another k-mer, ACCC, what is the probability, $P(\text{motif}=\text{ACCC}|\text{profile})=?$
 2. Both of the problems, *regulatory motif finding* and *clump finding for Oric* uses frequent words with approximate matches. Why is this algorithm more suitable for the latter problem compared to the first one?
 3. Suppose you have to find a k-mer which is a suspect motif in the following strings:
 ACCAGTCG
 TCGGTACG
 CAGTCGAT
 CAGTCAGT
 CGATCGAT
- Now simulate the following algorithm up to 2 iterations:
- a) Greedy Motif Search
 - b) Monte Carlo or Randomized Motif Search
 - c) Gibbs Sampler (for randomized generation, find out the probabilities and take the most probable one from the normalized probabilities)
 4. In Gibbs Sampling, after creating the profile matrix, the k-mer probability distribution does not add up to 1. Why?
 5. There is a parameter in Gibbs Sampler which is k. How can we fix this k? What happens with the increase and decrease of k?
 6. What problems might arise in Motif Finding with randomized algorithms if we have gene upstream regions with skewed distribution of nucleotides?

Genome Assembly

1. Why is de bruijn graphs preferred over overlap graphs in genome assembly problems?
2. Construct Overlap graph and de-bruijn graphs for the following DNA string:
 CGATCGATGTGTCGAT
 Assume k=3 and enumerate all kmers first. Also try to draw the de-bruijn graph for paired reads considering (k=3,d=1) reads.
3. Assume you have the following reads (k=10)
 GTGTGGGGCA
 GGGCAGCGAG
 GCGAGGTTTT
 You don't have all possible reads generated. Assume there is a full coverage with these 3 reads. Now, i) create a overlap graph ii) considering shorter k-mers (k=3) create a de-bruijn graph from these three reads. For both i) and ii) construct the superstring.
4. What happens to the de-bruijn graph if read length increases? Why we can not increase read length?
5. Does de-bruijn graph guarantee unique euler paths? How does that effect genome assembly? What are the solutions to this problem?

- How does creating shorter reads from original reads help in genome assembly?
- Explain contigs and bubbles and reason behind them.
- For each of the following graphs, find out if they have a euler path/cycle and hamiltonian path/cycle and if yes, find that.



Sequencing and Beyond

- You are given a data file with many low-quality reads (mean $Q < 15$). Describe how you would preprocess this data before assembly or variant calling.
- What is the purpose of the Phred+33 encoding scheme in FASTQ files?
- If the quality score of a base is 30, what is the estimated probability that the base is incorrect?
- Why do most sequencing reads have lower base quality toward the end?
- Why is it necessary to check the reverse complement of a read during genome mapping?
- A 100 bp read has high quality in the first 70 bases (Phred > 30) and drops to Phred < 15 in the last 30 bases. Would you prefer to align the full read or a trimmed read to the genome? Justify.

String Matching

- "Spaced K-mer indices can save space and time in read alignment" - do you support this statement? Please justify your answer.
- How exact matching algorithms can be used for approximate matching in an efficient way?
- For the following pattern and text, illustrate the working principle of the Boyer-Moore algorithm and show how much is saved using the Boyer-Moore algorithm.
Text: ACCGCGAGCGACGAGC
Pattern: CGAGC
- Repeat the last problem for the KMP algorithm.
- There are some pros and cons using Hash Tables and Multi-Maps for index data-structure. Explain each with examples.
- What are the ways to reduce the size of the index table? How does that affect the search efficiency?

Sequence Alignment

1. What are the applications of pairwise sequence similarity problem?
2. Why is local alignment more preferable than global alignment? What are the differences in dynamic programming formulation for these two problems?
3. Why do we need to address the affine gap penalties in sequence alignment problem? How does the dynamic programming formulation and runtime changes in this formulation?
4. Explain the application of multiple sequence alignment problem. Why we can not use dynamic programming to solve this problem?
5. What is BLAST? How does it save time while searching on a large database for sequence similarity?
6. Find the optimal global alignment of the following two sequences using -2 as the gap penalty, -1 as the mismatch penalty, and 2 as the score for a match. You have to use the Needleman-Wunsch algorithm, and show the corresponding dynamic programming (DP) table. In case there are multiple optimal alignments, find all of them. Please mark the paths, which correspond to the optimal alignments, in the DP table.
ATCCTGT
TCTCT
7. Consider the sequences and the scoring scheme mentioned in 6. Find the optimal local alignment(s) of these two sequences using Smith-Waterman algorithm. Show the DP table and mark the path(s) that corresponds to the alignment.
8. Now solve the global alignment version of the problem again for the affine gap penalty. Consider gap opening penalty -2 and gap extending penalty as -0.5.